Snippets

BA-MECH-24-4C

Control Engineering II

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Chapter 1

Text and Wording

1.1 Style (M019, M022)

The step response of the motor is recorded with an oscilloscope. The controller parameters are determined with the method described in Section 3.1. Important terms are *emphasised* in italics, not underlined.

1.2 Lists (M018)

- first item,
- second item.

1. The supply voltage is switched on.
2. The reference signal is applied.

1.3 Cross-References

The measured data are listed in Table 4.1 and plotted in Figure 5.5. The model follows from Equation (3.1) (see Chapter 3).

Chapter 2

Numbers and Units

- Quantity with unit: $12.5\,\mathrm{V}$, $3.2\,\mathrm{mA}$, $50\,\mathrm{mm}$
- Composite unit: $1.5\,\mathrm{m\,s^{-1}}$, $9.81\,\mathrm{m\,s^{-2}}$, $0.02\,\mathrm{N\,m\,s\,rad^{-1}}$
- Unit alone: The time is given in s, the speed in $\mathrm{rad\,s^{-1}}$.
- Pure number: 0.707, 1.2×10^{-3}, $12\,345.67$
- Uncertainty with $\pm$ (M031): $(100.0 \pm 1.2)\,\mathrm{g}$ or $(100.0 \pm 1.2)\,\mathrm{g}$
- Range and list: $0\,\mathrm{V}$ to $10\,\mathrm{V}$, $1\,\mathrm{Hz}$, $2\,\mathrm{Hz}$ and $5\,\mathrm{Hz}$
- Angle (no space, M029): $12.37°$, $12°22′12″$
- Percentage: $5\,\%$ overshoot

Chapter 3

Equations

3.1 Numbered Equation with Legend (M040, M042)

The plant is modelled as a first-order lag element, see Equation (3.1).

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{Ts + 1} \tag{3.1}$$

$G(s)$ transfer function of the plant, in $\mathrm{V\,V^{-1}}$

$Y(s)$ Laplace transform of the output signal, in $\mathrm{V\,s}$

$U(s)$ Laplace transform of the input signal, in $\mathrm{V\,s}$

K static gain, in $\mathrm{V\,V^{-1}}$

T time constant, in s

s Laplace variable, in $\mathrm{s^{-1}}$

3.2 Unnumbered Equation (M040)

$$\omega_\mathrm{n} = \sqrt{\frac{k}{m}}$$

3.3 Multi-Line Equations

$$y(t) = K\left(1 - \mathrm{e}^{-t/T}\right)$$
$$\dot{y}(t) = \frac{K}{T}\,\mathrm{e}^{-t/T}$$

3.4 Typesetting Rules (M043–M050)

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -ax + b\sin(\omega t), \qquad T_\mathrm{s} = 1\,\mathrm{ms}, \qquad x_i \text{ (i is an index variable, therefore italic)}$$

$$P = U \cdot I, \qquad F = ma$$

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{b}u, \qquad y = \mathbf{c}^\mathrm{T}\mathbf{x}$$

3.5 Symbols in Running Text (M041, M048)

The time constant T and the gain K are read from the step response.

3.6 Example Calculation (M099–M101)

The static gain is calculated from measurement 1 in Table 4.1:

$$K = \frac{\Delta y}{\Delta u} = \frac{4.20\,\mathrm{V} - 0.00\,\mathrm{V}}{2.00\,\mathrm{V} - 0.00\,\mathrm{V}} = 2.10\,\mathrm{V\,V^{-1}}$$

The remaining values are calculated analogously and listed in Table 4.1.

Chapter 4

Tables

4.1 Simple Table (M051–M057)

Table 4.1 lists the measured step responses.

Table 4.1: Measured input and output voltages of the step response experiment

Measurement	u/V	y/V
1	2.00	4.20
2	4.00	8.35
3	6.00	12.60

4.2 Table with Decimal Alignment and Composite Units

Table 4.2 shows the identified parameters.

Table 4.2: Identified parameters of the DC motor for different supply voltages

U/V	T/s	ω/(rad s^{-1})
6.0	0.105	98.4
12.0	0.112	197.0

4.3 Table Taken from a Source (M054, M069)

Table 4.3 gives the tuning rules according to [1].

Table 4.3: Ziegler–Nichols tuning rules for the step response method [1]

Controller	K_{P}	T_{I}	T_{D}
P	$1/a$	–	–
PI	$0.9/a$	$3L$	–
PID	$1.2/a$	$2L$	$L/2$

4.4 Full-Width Table with Text Columns

Table 4.4 lists the equipment used (M086, M088).

Table 4.4: List of equipment used for the experiment

Device	Description	Manufacturer/Type	Quantity
DC motor	brushed motor with gearbox and encoder	Maxon DCX 22 S	1
Oscilloscope	4 channels, $100\,\mathrm{MHz}$ bandwidth	Keysight DSOX1204G	1

Chapter 5

Figures

5.1 Single Figure (M058–M062, M064)

The experimental set-up is shown in Figure 5.1.

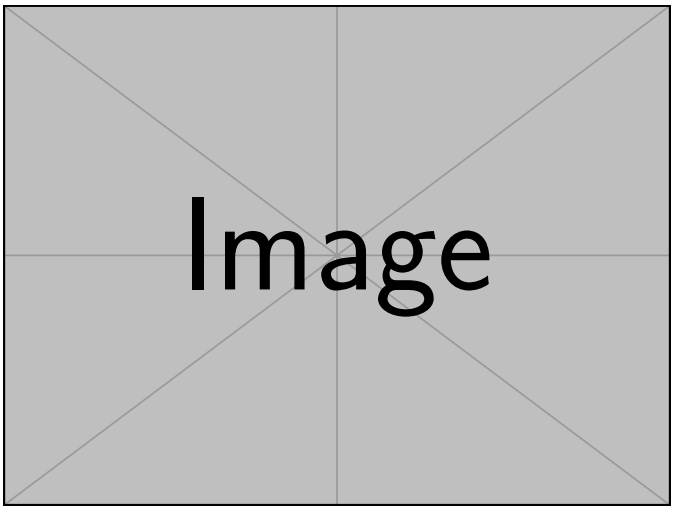


Figure 5.1: Experimental set-up of the DC motor test bench with all labelled components

5.2 Figure Taken from a Source (M069, M064)

Figure 5.2 is taken from the lecture notes [2].

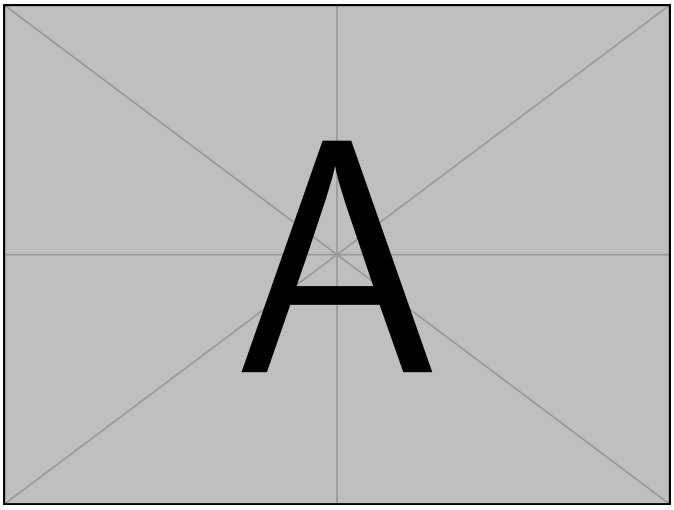


Figure 5.2: Block diagram of the standard control loop [2]

5.3 Two Figures Side by Side

Figures 5.3 and 5.4 compare both configurations.

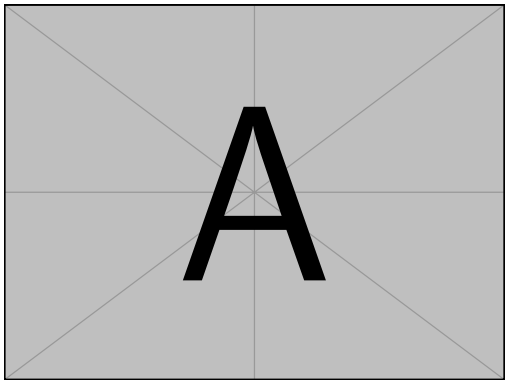


Figure 5.3: Test bench in configuration A

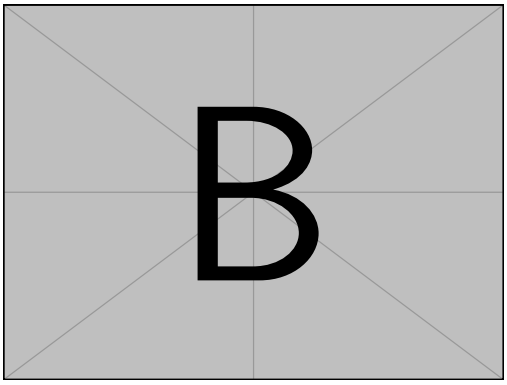


Figure 5.4: Test bench in configuration B

Tables 5.1 and 5.2 list the parameters of both tests.

Table 5.1: Parameters of test A

$K/(\mathrm{V\,V^{-1}})$	T/s
2.10	0.105

Table 5.2: Parameters of test B

$K/(\mathrm{V\,V^{-1}})$	T/s
2.08	0.112

5.4 Diagram with pgfplots (M012, M061–M068)

Figure 5.5 compares the measured and simulated step responses.

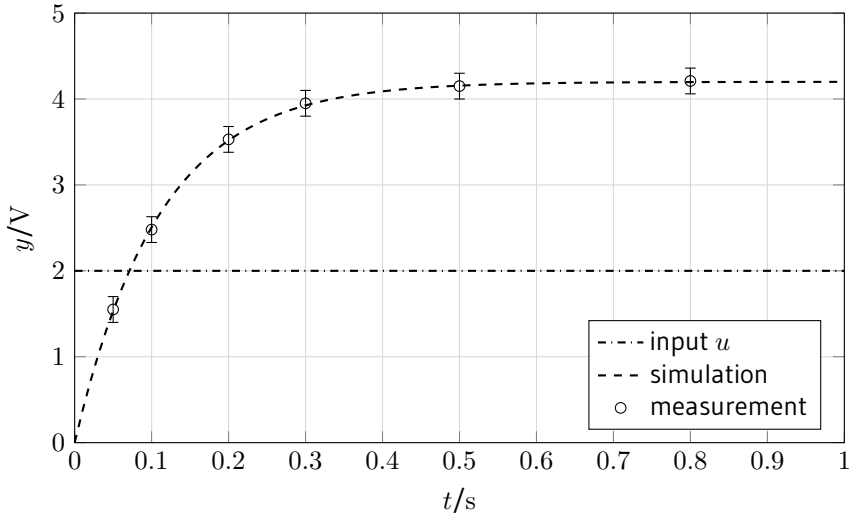


Figure 5.5: Comparison of measured (circles with error bars) and simulated (dashed) step response of the DC motor for an input step of $2\,\mathrm{V}$ (dash-dotted)

5.5 Diagram from MATLAB

The export code is given as a comment in the source file.

Chapter 6

Citations and Quotations

6.1 Citing (M069–M071, M108)

The step response method was originally proposed by Ziegler and Nichols and later revisited [1]. Several sources can be cited at once [2], [3]. The motor data are taken from the datasheet [4].

The following paragraph is based on [3]. A first-order lag element responds to a step input with an exponential approach to its final value. After one time constant, $63.2\,\%$ of the final value is reached.

6.2 Short Literal Quotation (M074)

According to the datasheet, the motor is "*exact wording from the source*" [4].

6.3 Long Literal Quotation (M075)

> Exact wording of the source, reproduced character by character [own addition in square brackets] and with omissions marked […] as shown here. [1]

6.4 Online Source (M081)

The figure was exported with the function `exportgraphics` [5].

Chapter 7

List of Symbols

Symbol	Description	Unit
K	static gain	$\mathrm{V\,V^{-1}}$
T	time constant	s
t	time	s
u	input voltage	V
y	output voltage	V
ω	angular velocity	$\mathrm{rad\,s^{-1}}$
g	gravitational acceleration, $9.81\,\mathrm{m\,s^{-2}}$	$\mathrm{m\,s^{-2}}$

Chapter 8

Code Listing

Listing 8.1: Identification of the time constant

```
y63 = 0.632 * y_end;          % 63.2 % of the final value
T   = t(find(y >= y63, 1));   % first sample above threshold
```

Bibliography

[1] K. J. Åström and T. Hägglund, "Revisiting the Ziegler–Nichols step response method for PID control", *Journal of Process Control*, vol. 14, no. 6, pp. 635–650, 2004. DOI: 10.1016/j.jprocont.2004.01.002

[2] Y. Kim, *Lecture notes control engineering II*, MCI – The Entrepreneurial School, Innsbruck, 2026.

[3] J. Lunze, *Regelungstechnik 1: Systemtheoretische Grundlagen, Analyse und Entwurf einschleifiger Regelungen*, 12th ed. Berlin: Springer Vieweg, 2020. DOI: 10.1007/978-3-662-60746-6

[4] Maxon Motor AG, *DCX 22 S datasheet*, Maxon Motor AG, Sachseln, 2024.

[5] The MathWorks, Inc. "Exportgraphics – save plot or graphics content to file", Accessed: Oct. 17, 2026. [Online]. Available: https://www.mathworks.com/help/matlab/ref/exportgraphics.html

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